amazon

SQL ANALYSIS PROJECT

Amazon Brazil

From Orders to Insights

Analyzing customers, sellers, products and operations to uncover key business insights for Brazil's fast-growing e-commerce market.



Customers
Who drives revenue?

Sellers
Who delivers?

Products
Which categories grow?

Orders
How is the platform performing?

Payments
Which method dominates?

Reviews
How satisfied are customers?

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Project Conclusion

Key learnings and overall impact from the analysis



Project Objective

Use SQL to analyze Amazon Brazil's e-commerce data to uncover insights on customers, products, sellers, delivery performance and reviews. The analysis helps understand key trends and support data-driven decision making.

Business Impact

- Helps identify **high-performing product categories** and growth opportunities.
- Provides visibility into **delivery performance** and customer satisfaction trends.
- Highlights **seller performance** differences across the platform.
- Supports actions to improve **customer engagement** and review participation.

SQL Skills Demonstrated

Data Retrieval
SELECT, WHERE, ORDER BY, LIMIT

Aggregation
COUNT, SUM, AVG, DISTINCT, GROUP BY, HAVING

Joins
INNER JOIN, LEFT JOIN

Date Functions
DATE_FORMAT, DATEDIFF

Window Functions
AVG() OVER(), ROW_NUMBER(), partition by

Subqueries / CTEs
Correlated subqueries, NOT EXISTS, WITH CTE

Conditional Logic
CASE WHEN

Data Analysis
Combining multiple tables to solve real business questions

Key Learnings

- Write efficient SQL queries for real-world business problems.
- Work with multiple tables and complex joins.
- Use aggregation, window functions and subqueries for deeper analysis.
- Translate business questions into SQL and interpret results.
- Derive meaningful insights to support data-driven decisions.

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Project Overview

TURNING TRANSACTION DATA INTO DECISIONS

Amazon Brazil is scaling rapidly in Latin America's largest e-commerce market. The operations and strategy team needs a clearer picture of how the platform is performing — which customers are driving revenue, which sellers are delivering, which product categories are growing, and where fulfillment is breaking down.

OBJECTIVE

Write SQL queries to turn Amazon Brazil's transactional data into actionable business insights.

DATA SCOPE

Customers	Sellers	Products	Orders	Payments	Reviews
Who drives revenue?	Who delivers?	Which categories grow?	How is the platform performing?	Which method dominates?	How satisfied are customers?





Q1 Total Orders Fulfilled by Each Seller State

Business Context

Amazon Brazil's logistics team is planning its next round of regional fulfillment center investments. Before committing budget, they need to know which states are generating the most order volume on the seller side — so infrastructure goes where the actual demand is.

Business Question

Find the **total number of orders fulfilled by each seller state.**

Key Insight (from query result)

SP has the highest order volume among seller states, followed by MG, PR, RJ and SC.

MySQL

SQL Query

MySQL Workbench

```sql
SELECT
    sellers.seller_state,
    COUNT(DISTINCT order_items.order_id) count_of_order
FROM
    sellers
    JOIN
    order_items USING (seller_id)
GROUP BY sellers.seller_state
ORDER BY count_of_order desc;
```

Query Result

Limit to 1000 rows

seller_state	count_of_order
SP	70188
MG	7930
PR	7673
RJ	4353
SC	3667



Q2 Cumulative Revenue by Product Category Over Time

Business Context

The growth team wants to see how revenue builds up within each product category across the order timeline. This helps them understand which categories hit their revenue targets early in the month versus which ones pick up later — useful for planning flash sales and targeted push notifications.

Business Question

For each product category, calculate the cumulative revenue generated as orders come in over time.

Key Insight (from query result)

The running total shows how revenue accumulates for each category over time, helping identify early versus late growth categories.

SQL Query

MySQL Workbench

```sql
WITH cte AS (
    SELECT
        products.product_category_name,
        DATE(orders.order_purchase_timestamp) AS order_date,
        SUM(order_items.price) AS revenue
    FROM products
        JOIN order_items USING (product_id)
        JOIN orders USING (order_id)
        GROUP BY products.product_category_name, order_date
)
SELECT
    product_category_name,
    order_date,
    revenue,
    SUM(revenue) OVER (PARTITION BY product_category_name ORDER BY order_date)
      AS running_total
FROM cte
ORDER BY order_date;
```

Query Result (Sample Output)

Limit to 1000 rows

product_category_name	order_date	revenue	running_total
NULL	2016-10-04	49.99	49.99
NULL	2016-10-08	15.9	65.89
NULL	2017-01-17	169	234.89
NULL	2017-01-20	349.99	584.88
NULL	2017-01-25	378.9	963.78

Q3 Most Used Payment Method and Average Order Value by Payment Type



Business Context

The finance and partnerships team is renegotiating contracts with payment providers. They need to know which payment method dominates the platform and whether high-value orders cluster around a specific payment type.

Business Question

Which payment method do customers use the most, and what is the average order value for each payment type?

Key Insight (from query result)

Credit card is the most used payment method with 76,505 orders and the highest average order value.

SQL Query

MySQL Workbench

```sql
with cte1 as (
    select
        order_id,
        payment_type,
        sum(payment_value) total_order_value
    from order_payments
    group by order_id, payment_type
),
cte2 as (
    select
        payment_type,
        count(order_id) payment_count,
        round(avg(total_order_value), 2) avg_order_value
    from cte1
    group by payment_type
)
select * from cte2
order by payment_count desc;
```

Query Result (Top 5)

Limit to 1000 rows

payment_type	payment_count	avg_order_value
credit_card	76505	163.94
boleto	19784	145.03
voucher	3866	98.15
debit_card	1528	142.66
not_defined	3	0.00

Q4 Customer Who Spent the Most Money Across All Their Orders



Business Context

Amazon Brazil wants to identify the highest-spending customer across all their orders. This helps in building premium loyalty programmes and targeted offers for high-value customers.

Business Question

Find the customer who has spent the most money across all their orders.

Key Insight (from query result)

Customer **1617b1357756262bfa56ab541c47bc16** has the highest total spending of **13,664.08** across all orders.

SQL Query

MySQL Workbench

```sql
SELECT
      o.customer_id,
     SUM(p.payment_value) AS total_spent
FROM orders o
JOIN order_payments p ON o.order_id = p.order_id
GROUP BY o.customer_id
ORDER BY total_spent DESC
LIMIT 1;
```

Query Result

Limit to 1000 rows

customer_id	total_spent
1617b1357756262bfa56ab541c47bc16	13664.08

Q5 Average Review Score by Product Category



Business Context

Amazon Brazil wants to analyze customer feedback by understanding the average review score for each product category.

Business Question

Find the average review score for each product category.

Key Insight (from query result)

SQL Query

MySQL Workbench

```sql
SELECT
    product_category_name,
    ROUND(AVG(review_score), 2) AS avg_review_score
FROM products
    JOIN order_items USING (product_id)
    JOIN order_reviews USING (order_id)
GROUP BY product_category_name;
```

Query Result

Limit to 1000 rows

product_category_name	avg_review_score
relogios_presentes	4.02
livros_tecnicos	4.37
informatica_acessorios	3.93
eletronicos	4.04
esporte_lazer	4.11



Q6 Total Orders by Customer and State

Business Context

Amazon Brazil wants to understand how many orders each customer has placed, along with the state they live in, to analyze customer distribution and ordering activity across states.

Business Question

Find the total number of orders placed by each customer, broken down by the state they live in.

Key Insight (from query result)

The output shows the total number of orders for each customer, along with the state they live in.

SQL Query

MySQL Workbench

```
1  SELECT
2      customer_id, customer_state, COUNT(order_id)
3  FROM customers
4      JOIN orders USING (customer_id)
5  GROUP BY customer_id , customer_state;
```

Query Result

Limit to 1000 rows

customer_id	customer_state	COUNT(order_id)
41ce2a54c0b03bf3443c3d931a367089	GO	1
d3e3b74c766bc6214e0c830b17ee2341	MG	1
3b6828a50ffe546942b7a473d70ac0fc	GO	1
7f8c8b9c2ae27bf3300f670c3d478be8	BA	1
d2b091571da224a1b36412c18bc3bbfe	SP	1

Q7 Sellers Who Never Fulfilled an Order



Business Context

Amazon Brazil wants to identify sellers who registered on the platform but have never fulfilled a single order. This helps in monitoring inactive sellers and improving seller engagement.

Business Question

Identify sellers who registered on the platform but have never fulfilled a single order.

Key Insight (from query result)

The output shows sellers who have zero delivered orders, meaning they have registered but have never fulfilled any order on the platform.

SQL Query

MySQL Workbench

```sql
select sellers.seller_id,
       count(case
        when orders.order_status = 'delivered' then 0
       end) delivered_orders_count
from sellers
left join order_items using (seller_id)
left join orders using (order_id)
group by sellers.seller_id
having delivered_orders_count = 0;
```

Query Result

Limit to 1000 rows

seller_id	delivered_orders_count
ad738d3fcf42655ed15c4960b7b6c601	0
02a2272692e13558373c66db98f05e2e	0
fc99f99385ae90f66f01733856fb88c9	0
4be2e7f96b4fd749d52dff41f80e39dd	0
37ceb404da208418c9032f0674e57804	0

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Q8 Top 5 Product Categories by Total Revenue



Business Context

The merchandising team uses category revenue to prioritize homepage placement and promotional budget for major sales events.

Business Question

Find the top 5 product categories by total revenue.

Key Insights (from query result)

- **beleza_saude** ranks first among the five categories shown.
- **relogios_presentes** is second.
- **informatica_acessorios** ranks fifth among these five.

SQL Query

MySQL Workbench

```sql
SELECT
    product_category_name, SUM(price) AS total_revenue
FROM
    products
     JOIN
     order_items USING (product_id)
GROUP BY product_category_name
ORDER BY total_revenue DESC
LIMIT 5;
```

Query Result

Limit to 1000 rows

product_category_name	total_revenue
beleza_saude	1258681.34
relogios_presentes	1205005.68
cama_mesa_banho	1036988.68
esporte_lazer	988048.97
informatica_acessorios	911954.32

Q9 Median Delivery Time (Days)



Business Context

The median delivery time represents the middle delivery duration and is less affected by unusually delayed orders than the average.

Business Question

Find the median delivery time (in days) between order placement and actual delivery.

Key Insight (from query result)

The median delivery time is **10.0000 days**, representing the middle delivery duration between order placement and actual delivery.

SQL Query

MySQL Workbench

```sql
with cte1 as (SELECT
      DATEDIFF(order_delivered_customer_date,
               order_purchase_timestamp) delivery_days
FROM
      orders where DATEDIFF(order_delivered_customer_date,
               order_purchase_timestamp) is not null),

cte2 as (select cte1.*,row_number() over(order by delivery_days) pos,
        count(delivery_days) over() n from cte1)
select avg(delivery_days) median_delivery_time from cte2 where pos in (ceil(n/2), ceil((n+1)/2));
```

Query Result

Limit to 1000 rows

median_delivery_time
10.0000



Q10 Products That Have Never Been Ordered

Business Context

Catalogue operations reviews products that have never received an order.

Business Question

Find all products that have never been ordered.

Key Insight

The query returned 0 rows. No products matched the condition in this result.

MySQL Workbench

```sql
SELECT
    p.product_id, p.product_category_name
FROM
    products p
WHERE
    NOT EXISTS (
        SELECT 1
        FROM order_items oi
        WHERE oi.product_id = p.product_id
    );
```

Query Result

Limit to 1000 rows

product_id	product_category_name

0 row(s) returned

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Q11 Sellers Above Average Order Volume



Business Context

The seller success team identifies sellers whose order count is above the platform average.

Business Question

Find sellers who have fulfilled more orders than the average seller on the platform.

Key Insight

The average seller order count is **32.3134**. Sellers returned by the query have counts greater than this average.

SQL Query

MySQL Workbench

```sql
WITH cte1 AS (
    SELECT seller_id, COUNT(DISTINCT order_id) AS count
    FROM sellers LEFT JOIN order_items USING (seller_id)
    GROUP BY seller_id
),
cte2 AS (
    SELECT seller_id, count,
           AVG(count) OVER() AS average_seller
    FROM cte1
)
SELECT * FROM cte2
WHERE count > average_seller;
```

Query Result

Limit to 1000 rows

seller_id	count	average_seller
001cca7ae9ae17fb1caed9dfb1094831	200	32.3134
002100f778ceb8431b7a1020ff7ab481	51	32.3134
004c9cd9d87a3c30c522c48c4fc07416	158	32.3134
00ee68308b45bc5e2660cd833c3f81cc	135	32.3134
00fc707aaaad2d31347cf883cd2dfe10	103	32.3134

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Q12 Highest Average Review Scores by State

Business Context

Customer reviews help the Amazon Brazil team understand customer satisfaction across different states for delivered orders.

Business Question

Find which Brazilian states have the highest average customer review score for orders delivered there.

Key Insight

SP has the highest average review score at **4.2462**, followed by **AP** at **4.2424** and **PR** at **4.2383**.

SQL Query

MySQL Workbench

```sql
SELECT
    customer_state, AVG(review_score) AS avg_customer_review
FROM
    customers
        JOIN
    orders USING (customer_id)
        JOIN
    order_reviews USING (order_id)
WHERE
    order_status = 'delivered'
GROUP BY customer_state
ORDER BY avg_customer_review DESC;
```

Query Result

Limit to 1000 rows

customer_state	avg_customer_review
SP	4.2462
AP	4.2424
PR	4.2383
AM	4.2207
MG	4.1927

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Q13 Customers Who Never Left a Review



Business Context

The post-purchase team identifies customers who placed orders but have no review, for review reminders.

Business Question

Identify customers who have placed orders but never left a review.

Key Insight

The query returned **768 rows**.

MySQL

SQL Query

MySQL Workbench

```sql
SELECT
    o.customer_id, r.review_id
FROM
    orders o
    LEFT JOIN
    order_reviews r ON o.order_id = r.order_id
WHERE
    r.review_id IS NULL;
```

Query Result

Limit to 1000 rows

customer_id	review_id
738b086814c6fcc74b8cc583f8516ee3	NULL
52006a9383bf149a4fb24226b173106f	NULL
4e7656e34357b93f14b40c6400ca3f6e	NULL
c3d8fc500d86b1c961ee144395c13a57	NULL
5baa82a2c45fa3220cb57d9881db3211	NULL

768 row(s) returned

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Q14 Month with the Highest Number of Orders

Business Context

Demand forecasting uses historical order volume to plan warehouse capacity, delivery staffing, and marketing campaigns.

Business Question

Find the month with the highest number of orders placed across the entire platform.

Key Insight

November 2017 had the highest order volume, with **7,544** orders.

MySQL

SQL Query

MySQL Workbench

```sql
SELECT
    DATE_FORMAT(order_purchase_timestamp, '%Y-%M') AS months,
    COUNT(order_id) AS order_placed
FROM
    orders
GROUP BY months
ORDER BY order_placed DESC
LIMIT 1;
```

Query Result

Limit to 1000 rows

months	order_placed
2017-November	7544

1 row(s) returned

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Key Takeaways

Summary of insights from 14 SQL analysis questions



01

Top Category by Revenue

beleza_saude generated the highest revenue **(1,258,681.34)**.

02

Median Delivery Time

The median delivery time across all delivered orders is **10 days.**

03

Sellers Above Average

Several sellers have fulfilled more orders than the platform average **(32.3134 orders).**

04

Highest Review Score by State

São Paulo (SP) has the highest average customer review score **(4.2462)**.

05

Customers Without Reviews

768 customers have placed orders but have never left a review.

06

Peak Ordering Month

November 2017 recorded the highest number of orders **(7,544)**.

Overall Insight

The analysis provides a clear view of customer behavior, product performance, seller activity, delivery efficiency and review trends on Amazon Brazil using SQL.

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Project Conclusion

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Business Impact

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